

MH1301 Discrete Mathematics

Handout 10: Graph Theory (4): Traversal of Trees, Spanning Trees

In our fourth handout on graph theory, we introduce the concept of trees on graphs. This handout is mostly on the terminologies used. This handout prepares us for the minimum spanning tree algorithms that we will cover in the next handout.

- Definition and Notations,
- Rooted Trees,
- Binary Trees,
- Spanning Trees.

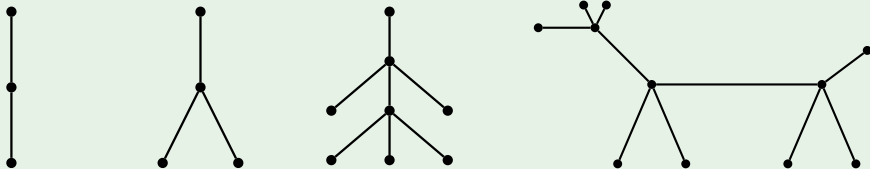
Trees

Definition 1 (Section 11.1)

- A **tree** is a connected graph that does not have a cycle as a subgraph.
- A **forest** is a graph that consists of **several trees**.
 - ▶ Forests are not necessarily connected.
 - ▶ The connected components of forests are trees.

*if one tree
→ also a forest*

Examples of trees



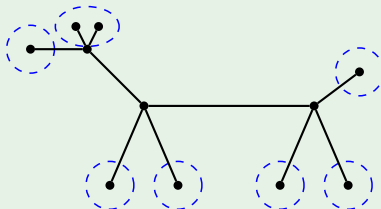
Question: What about forest?

Leaves

Leaves

In a tree T , a vertex with degree one is called a **leaf**.

Examples



Leaves

Tree Properties

Properties

- Trees and forests are **all planar**. *can represent the graph without edges crossing.*
- The planar representation of trees and forests has one region. *infinite region*
- Every tree (or forest) is bipartite. *no cycle* *→ no subgraph of odd cycle.*
- Every tree with n vertices, $n \geq 2$, must have **at least 2 leaves**.
- Removing a leaf v and the edge incident with v from any tree still gives a tree.

and many more...

Edges in a Tree

Theorem (Section 11.1)

A tree with n vertices has $n - 1$ edges.

Proof: Induction on the number of vertices n .

- Base: $n = 1$: isolated vertex; Claim is true.
- Hypothesis: assume the claim is true when $n = k$.
- Induction: Consider a tree T with $k + 1$ vertices. Let v be a leaf of T .
 - ▶ By the tree property, removing v and the edge e incident with it gives another tree $T - \{v\}$ with k vertices.
 - ▶ By hypothesis there are $k - 1$ edges in $T - \{v\}$.
 - ▶ Now add v and e back to $T - \{v\}$, we know there are k edges in T .

□

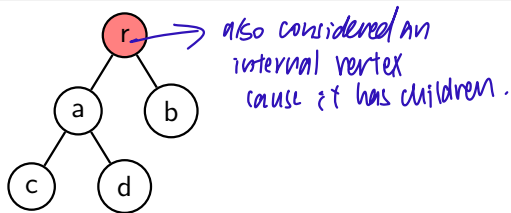
Alternative Definition for Tree

A connected graph with n vertices and $n - 1$ edges is a **tree**.

Rooted Tree

Definition 2 (Section 11.1)

A **rooted tree** is a tree in which one vertex is designated as the **root** and every edge is directed away from the root.



Leaves in a Rooted Tree

In a rooted tree, a vertex **that is not the root** and has degree one is called a **leaf**.

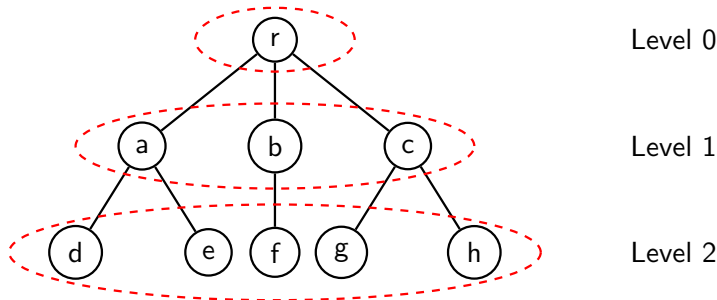
Vertices that have children are called **internal vertices**.

(Compare with leaf on a tree.)

Level and Height

Level and Height

- The **level** of a vertex u in a rooted tree is the length of the unique path from the root to this vertex. The level of the root is defined to be zero.
- The **height** of a rooted tree is the maximum of the levels of vertices.



Parent and Child

Parent and Child

Given a rooted tree T . If v is a vertex in T other than the root then

- the **parent** of v is the vertex u adjacent to v and one level **closer** from the root than v .
- the **children** of v are those vertices adjacent to v and one level **farther away** from the root than v .
 - ▶ in other words, if v is the parent of u , then u is called a **child** of v .

- The root of the tree has no parent, and is the only vertex with no parent. Every other vertex in the tree has a unique parent.
- A vertex in a tree may have 0, 1, or more than 1 children.

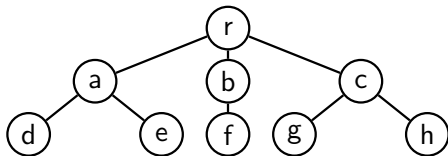
Ancestors and Descendants

- Let v be a vertex that is not the root. The **ancestors** of vertex v are the vertices in the path from the root to v , excluding v itself and including the root.
- The **descendants** of a vertex v are those vertices that have v as an ancestor.

Example

In the following tree:

- What are the parent and children of r , a , and d ?
- What are the ancestors and descendants of r , c , and h ?



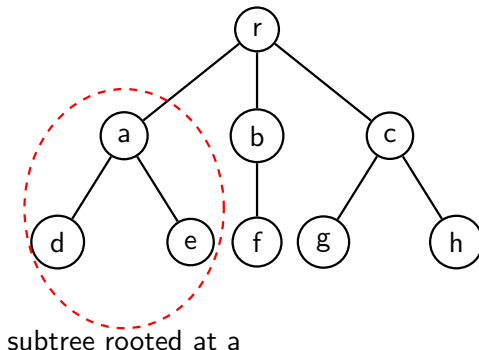
Solution:

- r has no parent. $Parent(a) = r$, $Parent(d) = a$.
- $Children(r) = \{a, b, c\}$, $Children(a) = \{d, e\}$, d has no children.
- r has no ancestor, $Ancestor(c) = \{r\}$, $Ancestor(h) = \{r, c\}$.
- $descendant(r) = \{a, b, c, d, e, f, g, h\}$, $descendant(c) = \{g, h\}$, h has no descendant.

Subtree

Subtree

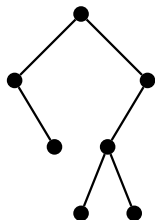
If u is a vertex of a tree, the **subtree** rooted at u is the subgraph of the tree consisting of u and all its descendants and all edges incident to these descendants.



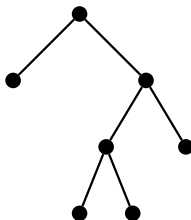
Binary Tree

Definition 3 (Case $m = 2$, Section 11.1)

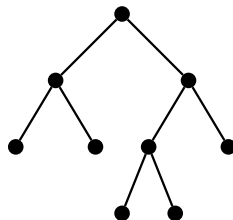
- A rooted tree is called a **binary** tree if every vertex has no more than 2 children.
- A rooted binary tree is called a **full binary tree** if every vertex other than the leaves has exactly two children.
- A rooted binary tree of height h is called a **balanced binary tree** if all leaves are at level h or $h - 1$.



binary tree



full binary tree



balanced binary tree

Binary Tree in Nature



Binary Tree

Theorem (Case $m = 2$, Section 11.1)

There are at most 2^h leaves in a binary tree T of height h .

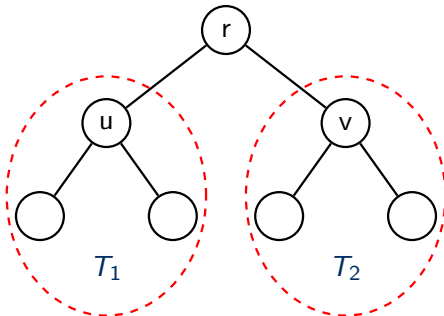
Proof: Apply mathematical induction on the height h .

- Basis: when $h = 1$, the root has at most 2 children.
- Hypothesis: Assume the claim is true when $h \leq k$.
- Inductive step: Consider $h = k + 1$. Let u and v be two children of the root and let T_1 and T_2 be the subtrees rooted at u and v , respectively.

Binary Tree

Then the height of T_1 and T_2 are at most k . Thus by the hypothesis, the number of leaves in T_1 and T_2 is at most 2^k each.

Hence, the number of leaves in T is at most $2 \cdot 2^k = 2^{k+1}$



[6] A tree with n edges has $n+1$ vertices.

[8] (T for true, F for false.) A forest with n vertices must have $n-1$ edges.
Answer: F

[9] (T for true, F for false.) The chromatic number of a tree is always two or less. Answer: T

[10] (T for true, F for false.) An undirected simple graph with n vertices and $n-1$ edges is always connected. Answer: F

[11] (T for true, F for false.) Let T be a tree. After removing any vertex v from T and the edges incident with v , the remaining graph $T - \{v\}$ is still a tree.
Answer: F

[12] A rooted tree with n vertices has $n-1$ edges.

[13] (T for true, F for false.) For a rooted tree, the root is always an internal vertex. Answer: F

[14] (T for true, F for false.) Leaves of a rooted tree do not have children.
Answer: T

[15] (T for true, F for false.) A rooted tree of height h and having 2^h vertices is a full binary tree. Answer: F

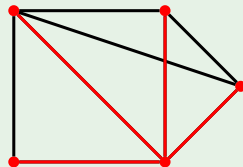
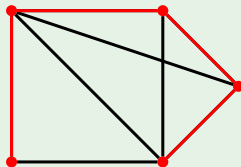
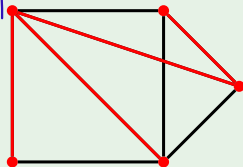
[16] (T for true, F for false.) Let T be a rooted tree. Suppose $v \in T$ has degree 2, then T contains at least two leaves. Answer: F

Spanning Trees

Definition 1 (Section 11.4)

Let G be a connected simple graph. A **spanning tree** of G is a subgraph of G that is a tree containing all vertices of G .

f *starting point*
Examples of spanning trees



Edge-Weighted Graphs

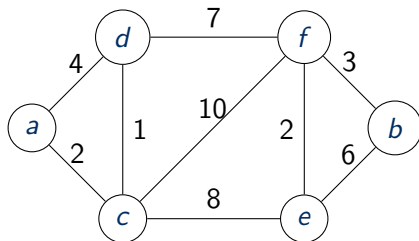
- In **edge-weighted graphs** every edge is associated with a number.
- This number is usually called the **edge weight**.
- The weight can mean a lot of things
 - ▶ distance between two vertices (e.g., traveling time)
 - ▶ cost of using the connection/edge
 - ▶ cost of maintaining the connection
 - ▶ capacity of the edge (think of pipes transporting water)
 - ▶ failure probability
 - ▶ etc.
- Many interesting graph problems concern edge-weighted graphs
 - ▶ minimum spanning tree problem
 - ▶ shortest path problem
 - ▶ etc.

Edge-Weighted Graphs

Edge-Weighted Graphs

An **edge-weighted** graph is a triple $G = (V, E, W)$, where V contains the vertices, E the edges ((V, E) is a graph), and $W : E \mapsto \mathbb{R}$ is a function that maps edges to weights.

- We will simply use the term “graph” from now on (whether weights are there or not will be obvious.)
- Sometimes edge weights will be required to be non-negative.

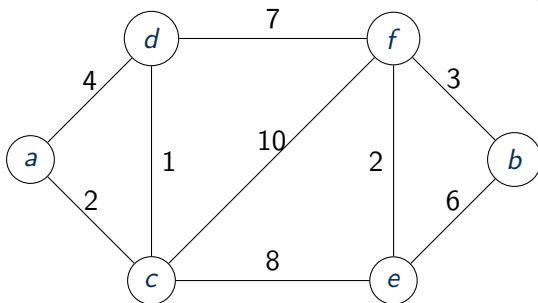


Example

A company plans to build a communications network connecting its six computer centers. Which links should be made to ensure that

- there is a path between any two computer centers
- the total cost of the network is minimized?

} you want a tree where the sum of the weights is the smallest.



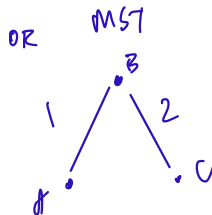
Minimum Spanning Trees

Definition 1 (Section 11.5)

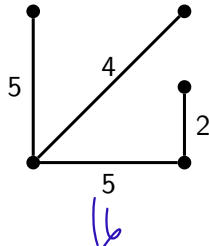
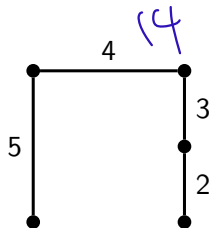
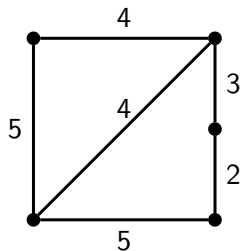
Given an edge-weighted graph G , a **minimum spanning tree** of G is a spanning tree that has the **smallest possible sum** of weights of its edges.

not unique

still need to
connect all
the vertices

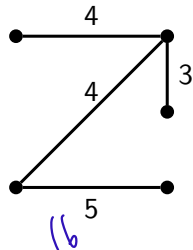
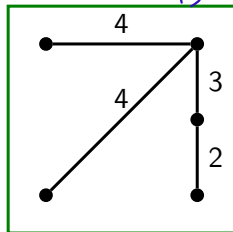


Example



all 5 vertices connected.

MST



Minimum Spanning Tree

Question

How to find a minimum spanning tree in an edge-weighted graph?

Stay Tuned.

Reading Assignment

- Rosen (7th edition):
 - ▶ Chapter 11.1. Introduction to Trees
 - ▶ Chapter 11.4. Spanning Trees